DIABETES AND HEALTHCARE DATA ANALYSIS

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INTRODUCTION

- Diabetes is the name of a **Chronic Disease** which refers to the **higher blood sugar level** more than the normal level required in our body.
- These are categorized as Normal, Pre-Diabetes and Diabetes.
- **The tests are:**

1)Fasting Blood Sugar(After not eating for 8-12hrs)

2)Post-Meal(2hrs After eating)

3)HbA1c Test(3-month blood sugar avg)

- **Main Types of Diabetes:**

1) Type 1 diabetes: Your immune system attacks the cells in your pancreas that make insulin.

2)Type 2 diabetes: Your body does not make enough insulin or resists how it works. This is the most common type.

3)Gestational diabetes: This type develops in some women during pregnancy.

In this Dataset The Value 0 is Normal and 1 is Pre-Diabetes or Diabetes

World Diabetes Day

November 14

"Scientific data suggest positive relationships between a vegetarian diet and reduced risk for obesity, coronary artery disease, hypertension, diabetes mellitus, and some types of cancer."



EXPLORATORY DATA ANALYSIS(EDA)

- **Initially in this Dataset there was 253680 data's in which I found that 24206 data's as Duplicates and I Cleaned that Data using the Pandas library in Python and the Python Code,i.e:df.drop_duplicates(inplace=True) and got it reduced to 229474 Data.**
- **By analyzing the dataset I found that only 15% of the people are Pre-Diabetic or Diabetic according to this Dataset.**
- **Top Diabetes Factors in this dataset are:**

1) GenHlth

2) HighBP

3) DiffWalk

4) BMI

5) HighChol

- **Top Preventive Factors in this dataset are:**

1) Income

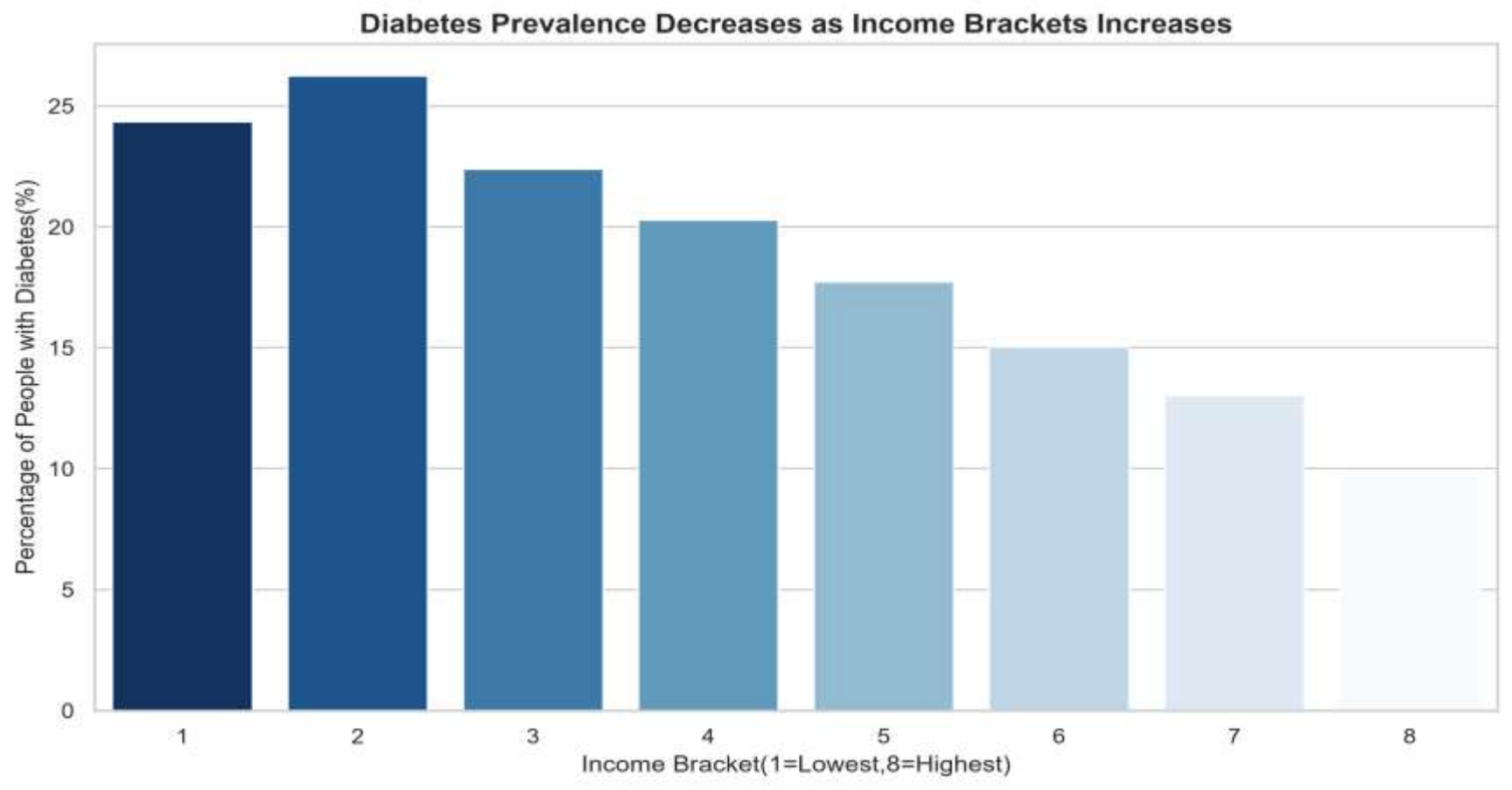
2) Education

3) PhysActivity

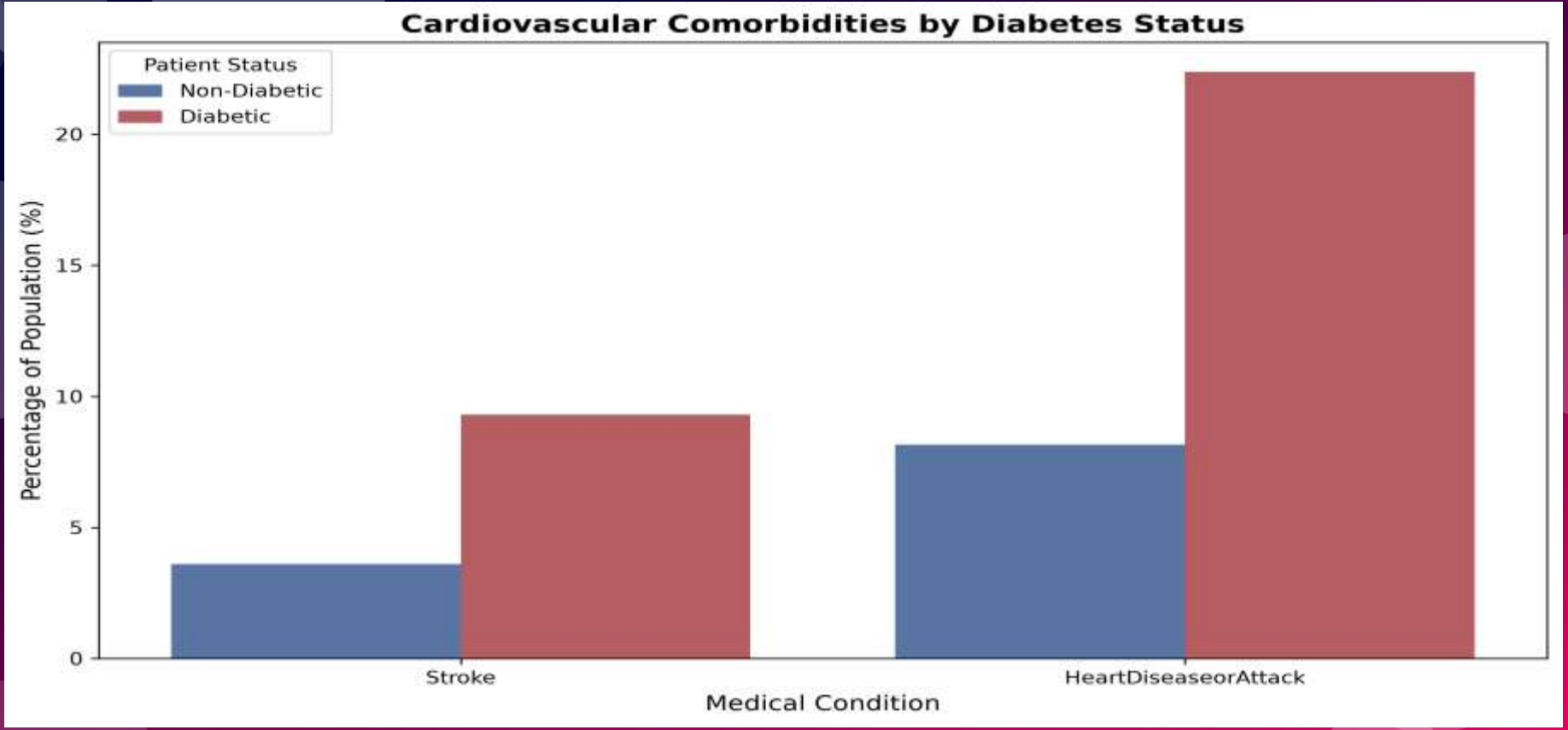
4) Veggies

5) Fruits

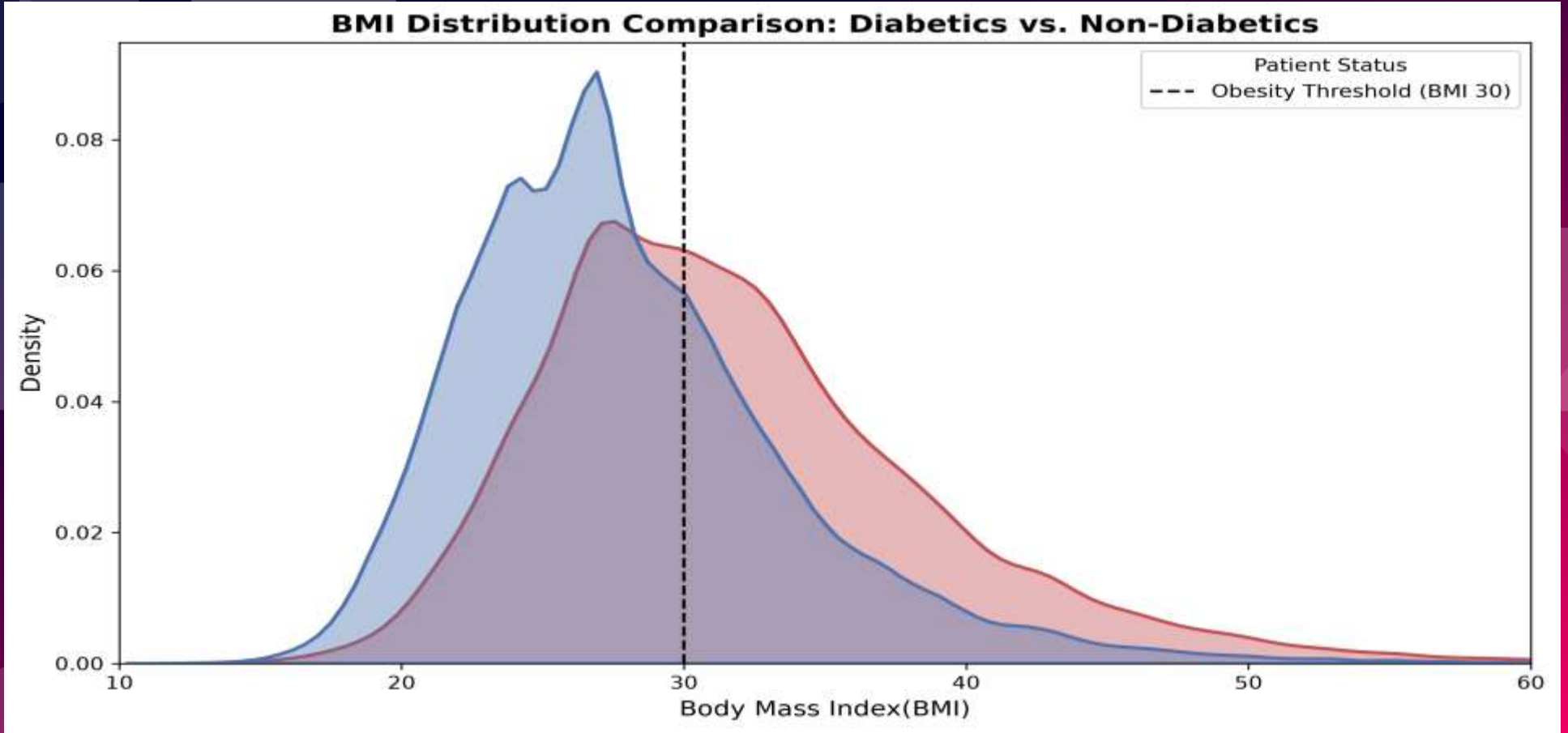
RESULT


Diabetes Prevalence Decreases as Income Brackets Increases
Percentage of People with Diabetes(%)
0
5
10
15
20
25
1
2
3
4
5
6
7
8
Income Bracket(1=Lowest,8=Highest)

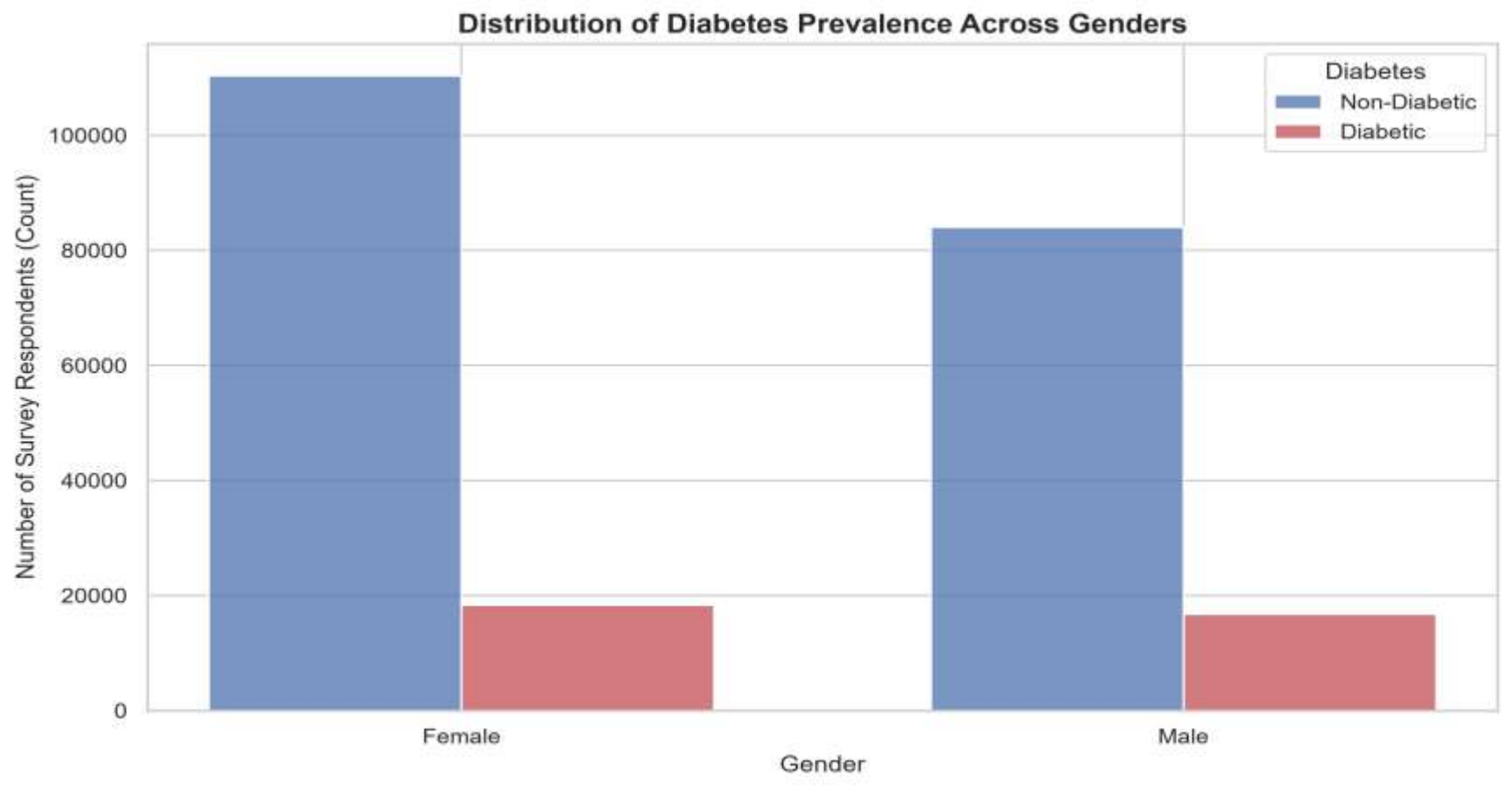
RESULT


Cardiovascular Comorbidities by Diabetes Status
Patient Status
Non-Diabetic
Diabetic
Percentage of Population (%)
0
5
10
15
20
Stroke
HeartDiseaseorAttack
Medical Condition

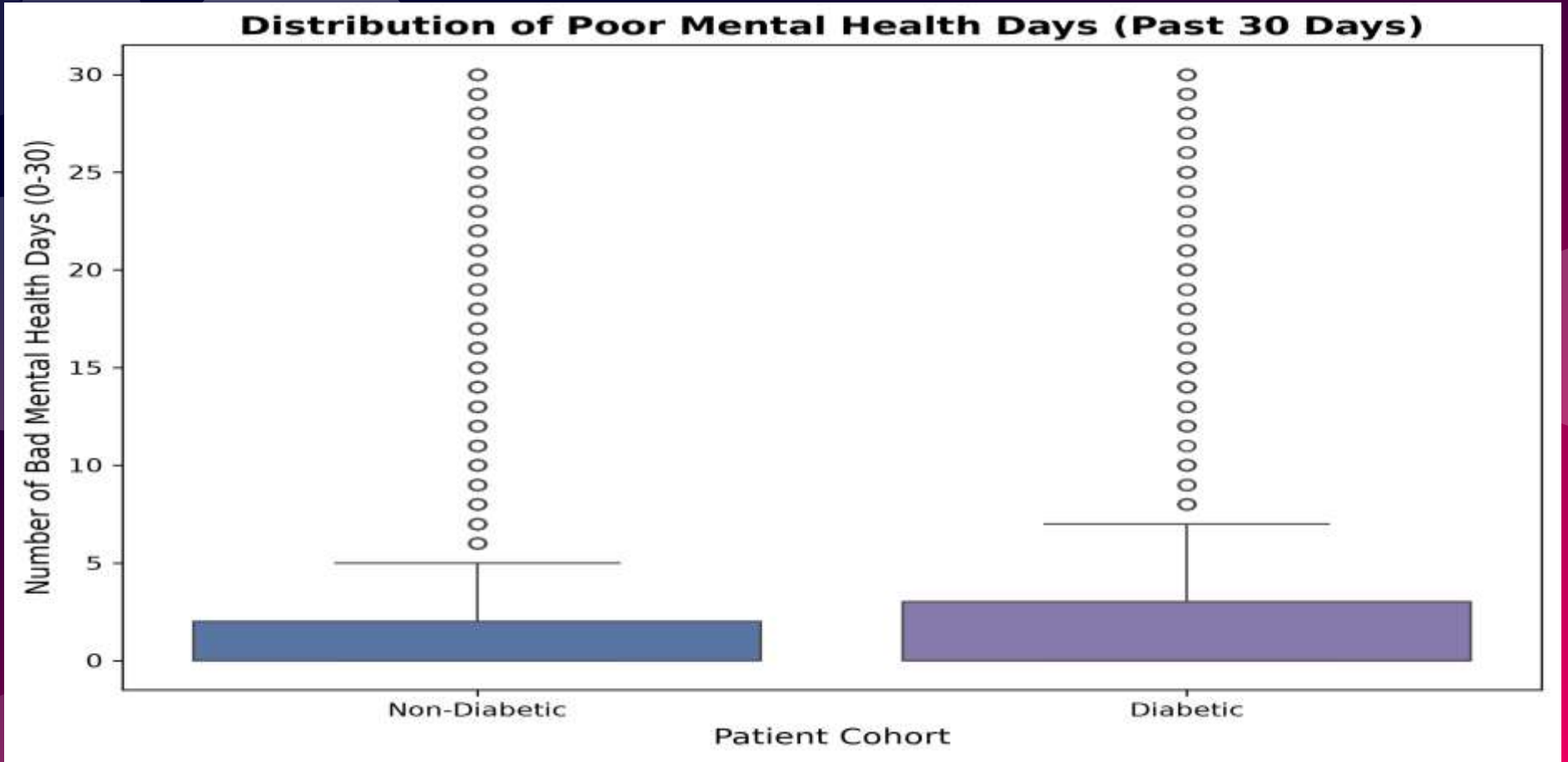
RESULT


BMI Distribution Comparison: Diabetics vs. Non-Diabetics
Patient Status
Obesity Threshold (BMI 30)
Density
0.00
0.02
0.04
0.06
0.08
10
20
30
40
50
60
Body Mass Index(BMI)

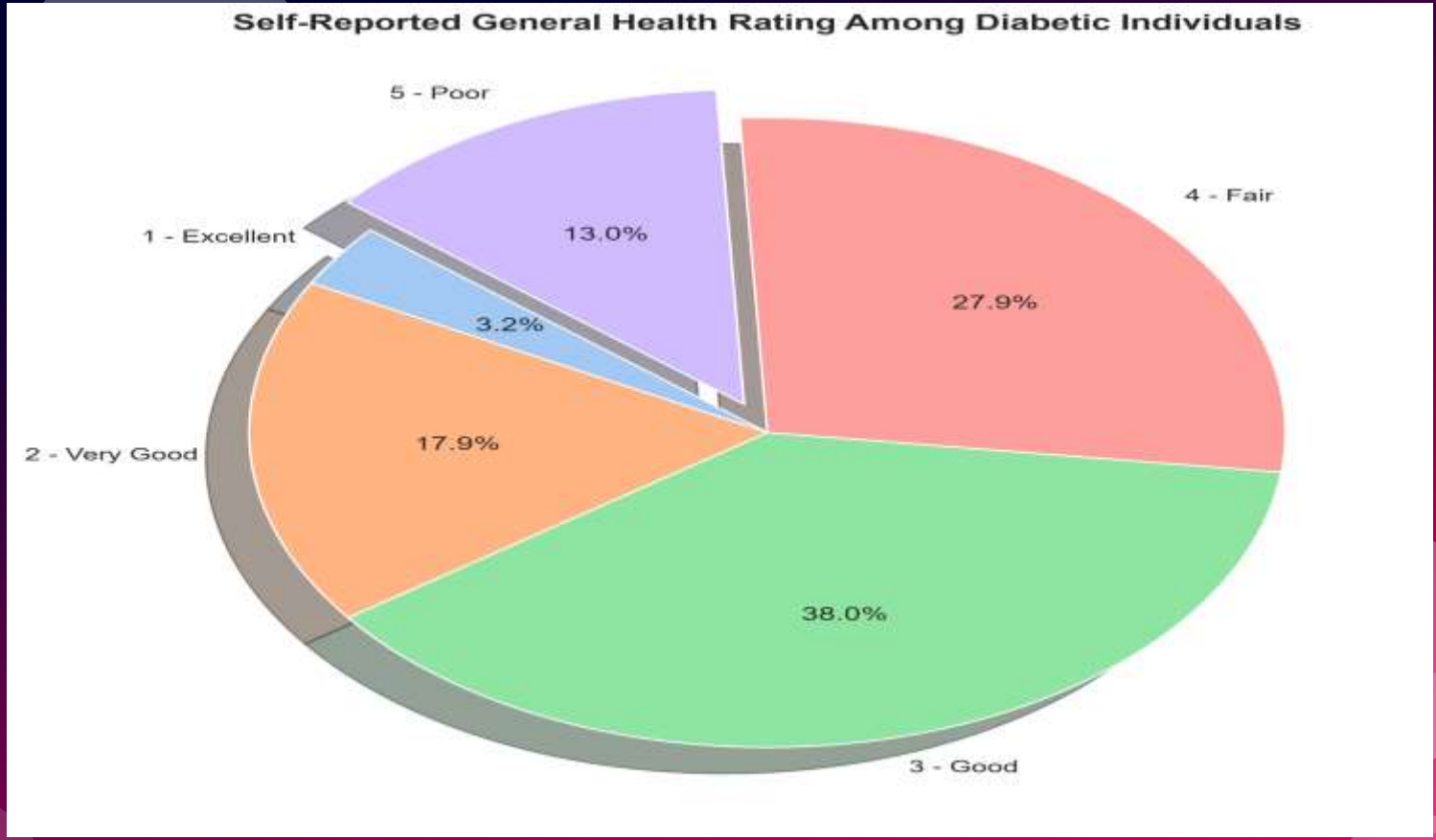
RESULT


Distribution of Diabetes Prevalence Across Genders
Diabetes
Non-Diabetic
Diabetic
Number of Survey Respondents (Count)
100000
80000
60000
40000
20000
0
Female
Male
Gender

RESULT


Distribution of Poor Mental Health Days (Past 30 Days)
Number of Bad Mental Health Days (0-30)
30
25
20
15
10
5
0
Non-Diabetic
Diabetic
Patient Cohort

RESULT


Self-Reported General Health Rating Among Diabetic Individuals
5 - Poor
13.0%
4 - Fair
27.9%
1 - Excellent
3.2%
2 - Very Good
17.9%
38.0%
3 - Good

CONCLUSION

- **Chronic, Long-Term Complications(Most Common):**

1) Heart Disease and Stroke

2) Kidney Failure

3) Severe Infections

- **Acute, Sudden Emergencies(Immediate Danger):**

1) Diabetic Keto-acidosis (DKA)

2) Hyperosmolar Hyperglycemic State (HHS)

3) Severe Hypoglycemia (Low Blood Sugar)

- So, these are some of the causes for death due to damage to the body's organs.
- To Avoid this Chronic disease and it's Complications we should take preventive measures like, doing some kind of Physical Activity,Eating Fruits and Vegetables and Not-Drinking Alcohol or Smoking.
- We Should Take the Diagnostic tests at least once every 3 months so, That we can prevent ourselves from any further Disease or Complications and to be able to take the required Treatment to cure it at the Earliest if we are Diagnosed with it.

THANK YOU